

2. Discuss Message Authentication protocol

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1. Definition:

A Message Authentication Protocol is a method used to verify that a message is sent by a legitimate sender and has not been altered during transmission. It provides authentication and data integrity.

2. Need for Message Authentication Protocol:

- To prevent message modification
- To avoid masquerade (fake sender) attacks
- To prevent replay attacks
- To ensure data integrity and authenticity

3. Types of Message Authentication Protocols:

a) Message Authentication Code (MAC):

- Uses a secret key shared between sender and receiver
- Generates a MAC value using message and key
- Provides integrity and authentication

b) Hash Function Based Protocol:

- Uses hash functions like MD5, SHA
- Produces a fixed-size message digest
- Does not provide authentication unless combined with a key

c) HMAC (Hash-based Message Authentication Code):

- Combines hash function and secret key
- More secure than simple hash functions

d) Digital Signature Protocol:

- Uses public key cryptography
- Sender signs message using private key
- Receiver verifies using public key
- Provides authentication, integrity, and non-repudiation

e) Authentication using Encryption:

- Entire message is encrypted using a key
- Only authorized receiver can decrypt and verify

4. Working of Message Authentication Protocol:

1. Sender creates a message (M)
2. Applies authentication technique (MAC / Hash / Signature)
3. Sends message along with authentication data
4. Receiver recomputes authentication value
5. Compares with received value
6. If values match, message is accepted; otherwise, rejected

5. Advantages:

- Ensures secure communication
- Detects message tampering
- Prevents unauthorized access

6. Conclusion:

Message Authentication Protocols are essential in cryptography to ensure secure, reliable, and trustworthy communication by protecting against various network attacks.



